



Ansys Fluent Simulation Report

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System Information

Application	Fluent
Settings	2d, double precision, pressure-based, dynamic mesh, Scale Adaptive Simulation, transient
Version	25.2.0-10204
Source Revision	5eecd5d865
Build Time	Jun 16 2025 10:44:41 EDT
CPU	AMD Ryzen 7 6800H with Radeon Graphics
OS	Windows

Geometry and Mesh

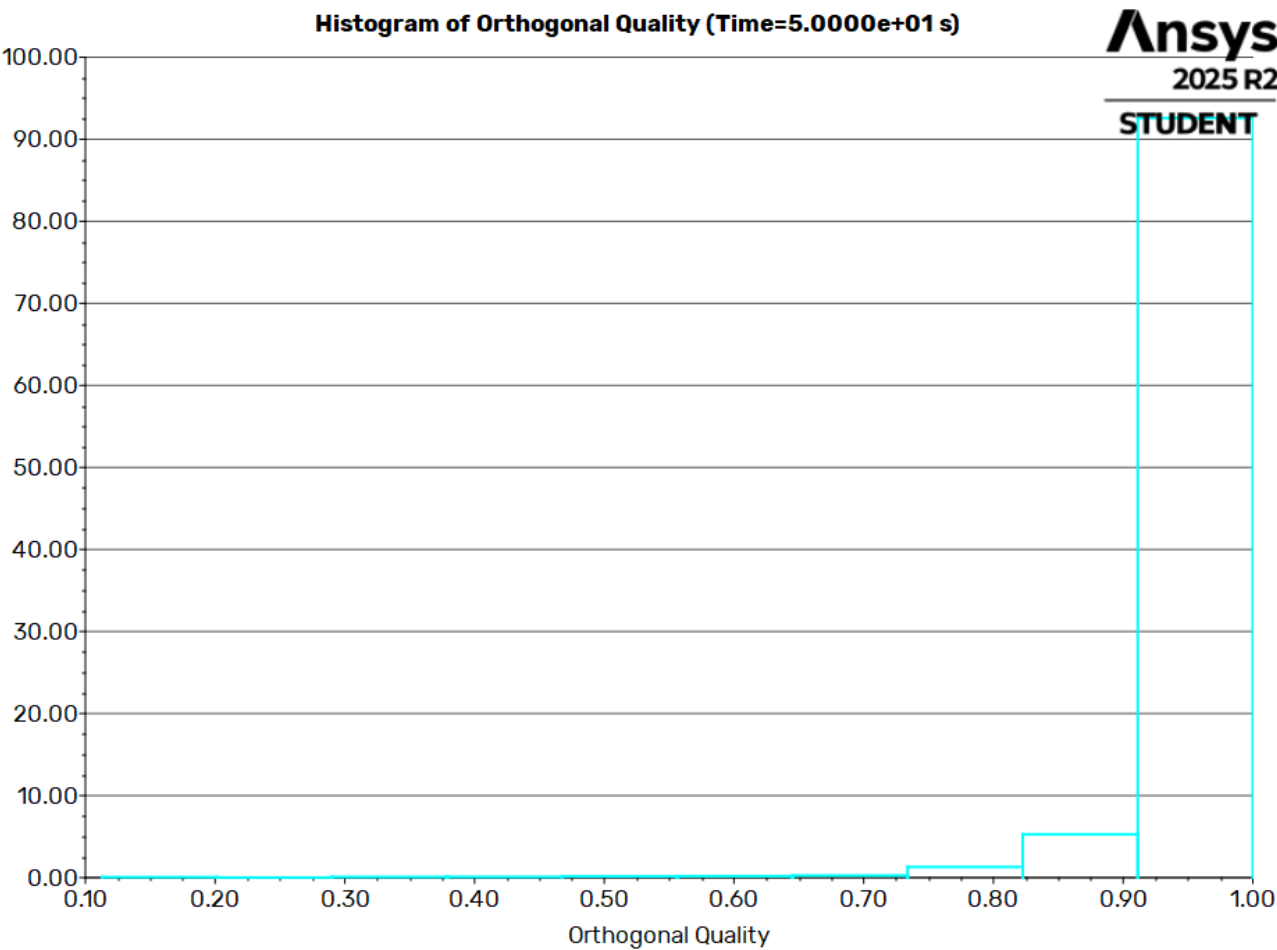
Mesh Size

Cells	Faces	Nodes
5840	8882	3043

Mesh Quality

Name	Type	Min Orthogonal Quality	Max Aspect Ratio
fluid	Tri Cell	0.1127685	10.245138
solid	Tri Cell	0.76980036	3.1622777

Orthogonal Quality



Simulation Setup

Physics

Models

Model	Settings
Space	2D
Time	Unsteady, 1st-Order Implicit
Viscous	SAS model

Material Properties

– Fluid	
– water-liquid	
Density	998.2 kg/m^3
Viscosity	0.001003 kg/(m s)
– air	
Density	1.225 kg/m^3
Viscosity	1.7894e-05 kg/(m s)
– Solid	
– aluminum	
Density	2719 kg/m^3

Cell Zone Conditions

– Fluid	
– fluid	
Material Name	water liquid
Specify source terms?	no
Specify fixed values?	no
Frame Motion?	no

Mesh Motion?	yes
Relative To Cell Zone	-1
Moving Mesh Rotation Speed [rad/s]	0
Moving Mesh X-Velocity Of Zone [m/s]	0
Moving Mesh Y-Velocity Of Zone [m/s]	0
Moving Mesh User Defined Zone Motion Function	none
Laminar zone?	no
Porous zone?	no
– Solid	
– solid	
Material Name	aluminum
Frame Motion?	no
Mesh Motion?	no
Solid Motion?	yes
Relative To Cell Zone	-1
Solid Body Rotation Speed [rad/s]	0
Solid Motion X-Velocity Of Zone [m/s]	0
Solid Motion Y-Velocity Of Zone [m/s]	0
Solid Motion User Defined Solid Motion Function	none

Boundary Conditions

– Wall	
solid_walls-shadow	wall
– solid_walls	
Wall Motion	Stationary Wall
Shear Boundary Condition	No Slip
Wall Roughness Height [m]	0
Wall Roughness Constant	0.5
– outer_walls	
Wall Motion	Stationary Wall
Shear Boundary Condition	No Slip

Wall Roughness Height [m]	0
Wall Roughness Constant	0.5

Reference Values

Area	1 m ²
Density	1.225 kg/m ³
Depth	1 m
Enthalpy	0 J/kg
Length	1 m
Pressure	0 Pa
Temperature	288.16 K
Velocity	1 m/s
Viscosity	1.7894e-05 kg/(m s)
Ratio of Specific Heats	1.4
Yplus for Heat Tran. Coef.	300
Reference Zone	fluid

Solver Settings

– Equations	
Flow	True
Turbulence	True
– Numerics	
Absolute Velocity Formulation	True
– Unsteady Calculation Parameters	
Number of Time Steps	100
Time Step Size [s]	0.5
Max Iterations/Time Step	40
– Under-Relaxation Factors	
Pressure	0.3
Density	1
Body Forces	1
Momentum	0.7

Turbulent Kinetic Energy	0.8
Specific Dissipation Rate	0.8
Turbulent Viscosity	1
– Pressure-Velocity Coupling	
Type	SIMPLE
– Discretization Scheme	
Pressure	Second Order
Momentum	Bounded Central Differencing
Turbulent Kinetic Energy	Second Order Upwind
Specific Dissipation Rate	Second Order Upwind
– Solution Limits	
Minimum Absolute Pressure [Pa]	1
Maximum Absolute Pressure [Pa]	5e+10
Minimum Static Temperature [K]	1
Maximum Static Temperature [K]	5000
Minimum Turb. Kinetic Energy [m ² /s ²]	1e-14
Minimum Turb. Kinetic Energy [m ² /s ²]	1e-07
Minimum Spec. Dissipation Rate [s ⁻¹]	1e-12
Minimum Spec. Dissipation Rate [s ⁻¹]	1e-20
Maximum Turb. Viscosity Ratio	100000

Run Information

Number of Machines	1
Number of Cores	2
Case Read	7.157 seconds
Iteration	39.247 seconds
AMG	8.20299 seconds
Virtual Current Memory	1.09753 GB
Virtual Peak Memory	1.21539 GB
Memory Per M Cell	77.0347

Solution Status

Flow Time: 50

Time Step: 100

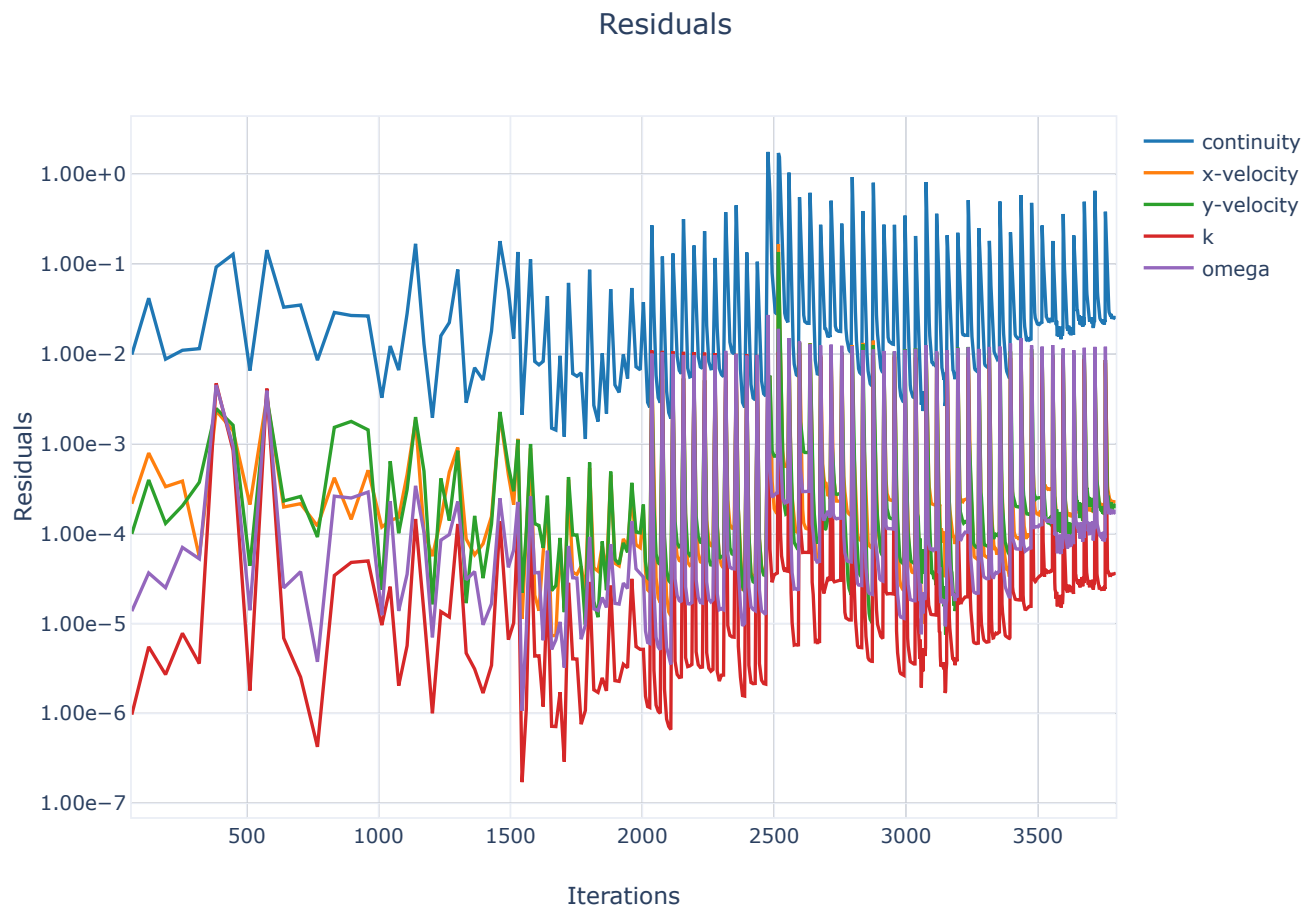
	Value	Absolute Criteria	Convergence Status
continuity	0.02651931	0.001	Not Converged
x-velocity	0.0002259744	0.001	Converged
y-velocity	0.0002154885	0.001	Converged
k	3.634933e-05	0.001	Converged
omega	0.0001857765	0.001	Converged

Report Definitions

delta-time	0.5	s
iters-per-timestep	40	
flow-time	50	s

Plots

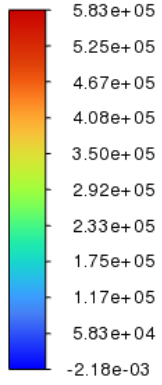
Residuals



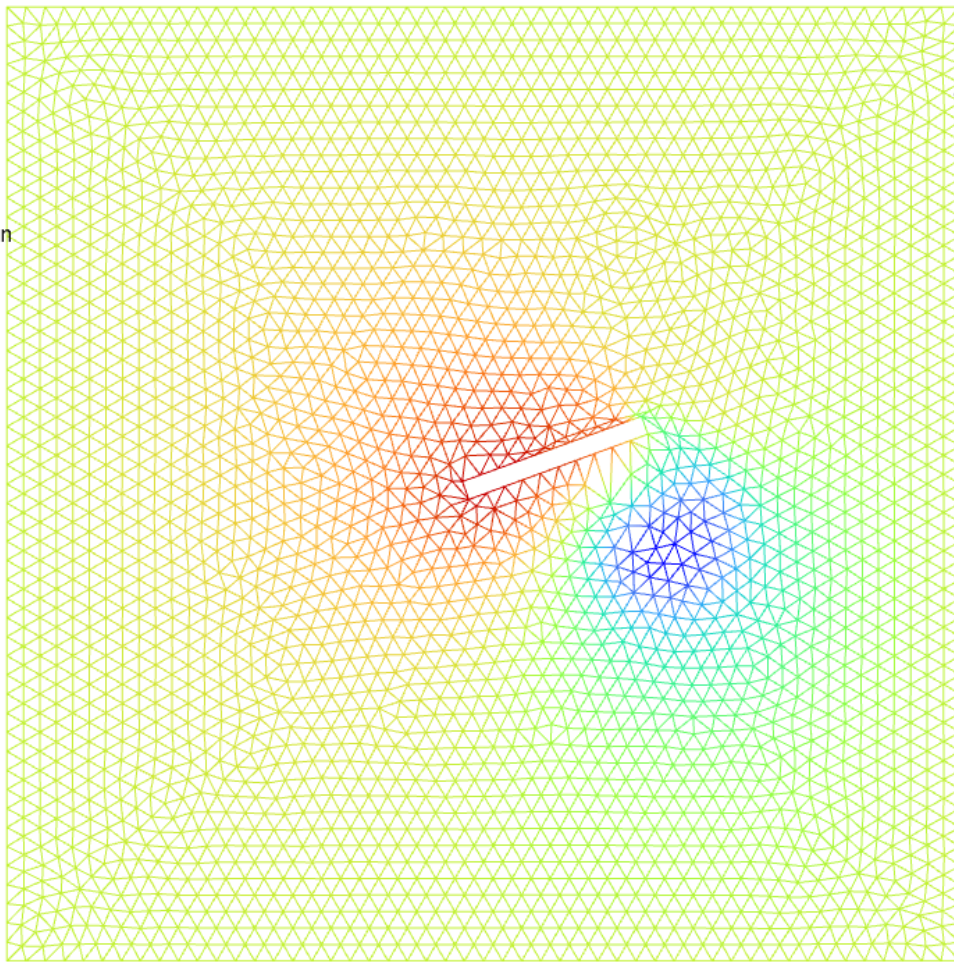
Contours

contour-1

Stream Function
[kg/s]



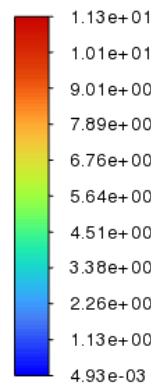
contour-1



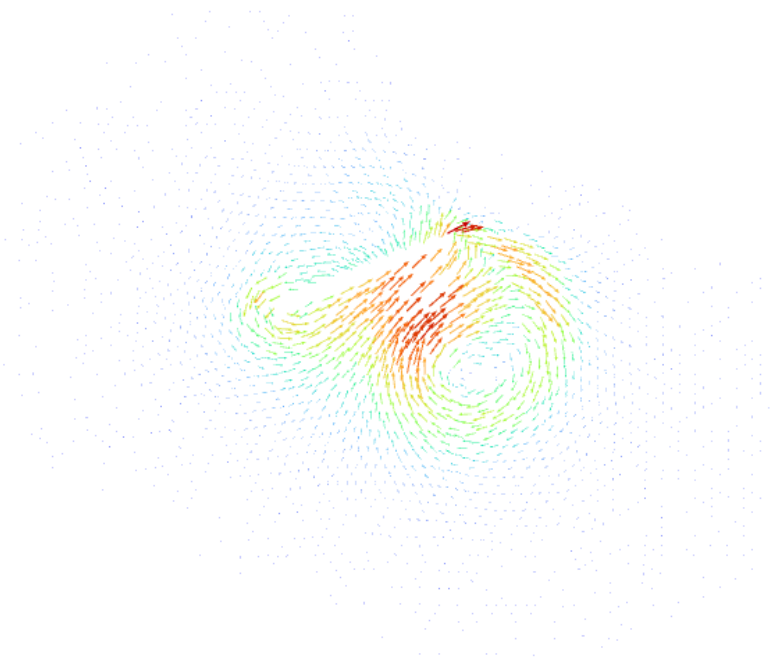
Vectors

vector-1

Velocity Magnitude
[m/s]



vector-1



Pathlines

pathlines-41

